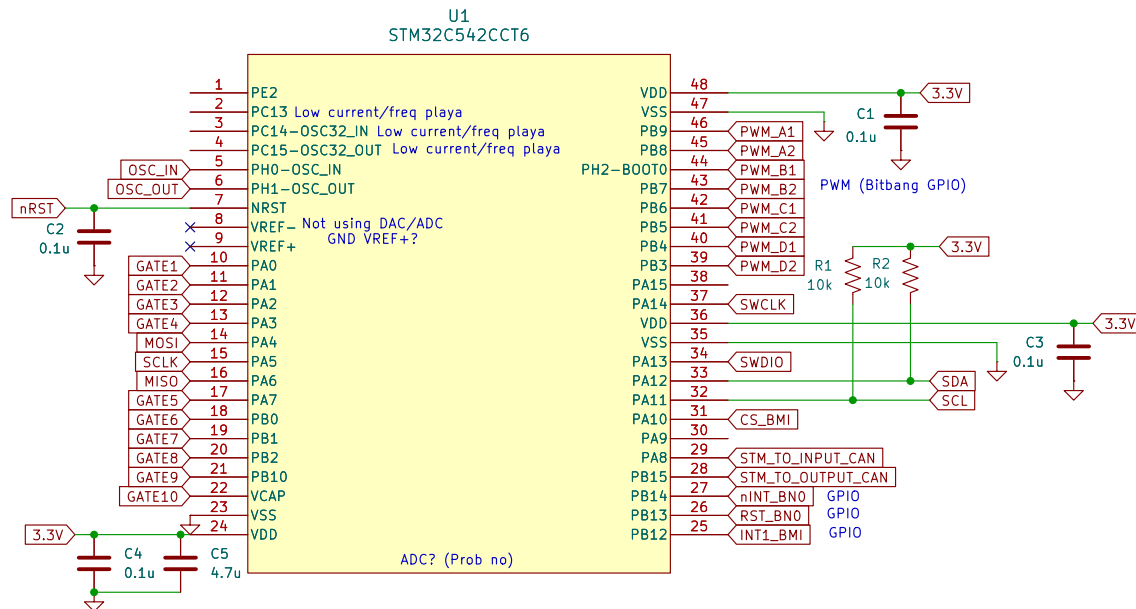


MCU Connections



Categorized Designs

MOSFET_driver



File: untitled.kicad_sch

Sensors_Telemetry



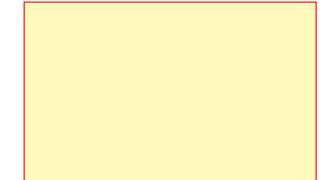
File: Sensors_Telemetry.kicad_sch

Connector_Signals



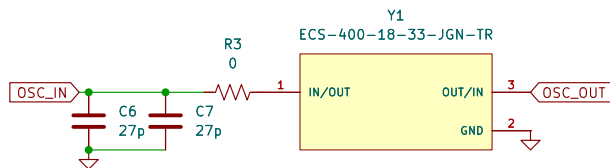
File: connector_signals.kicad_sch

Voltage_Conv-Protect-Monitoring



File: Voltage_conv_prot.kicad_sch

Crystal Oscillator



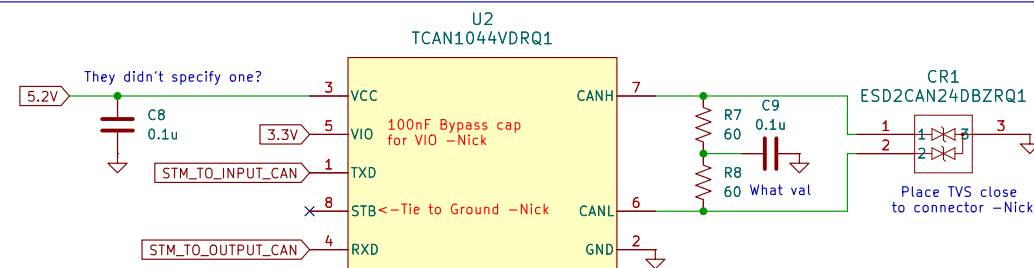
MCU:

- add debug LED to STM GPIO
- break out a few GPIO STM pins to a pin header
- why do gpio bitbang for PWM? Cant we just use the STM's timer channels?
- add no connect (NC) to the unused STM pins

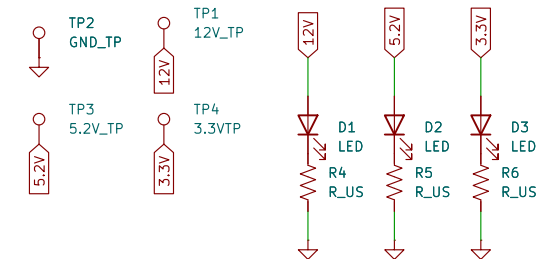
General:

- testpoints and debugging:
 - break out CAN signals to a header for debugging
 - break out SPI

CAN Transceiver



Test Points, LEDs



Resistor Values? - Nick

Purdue ROV

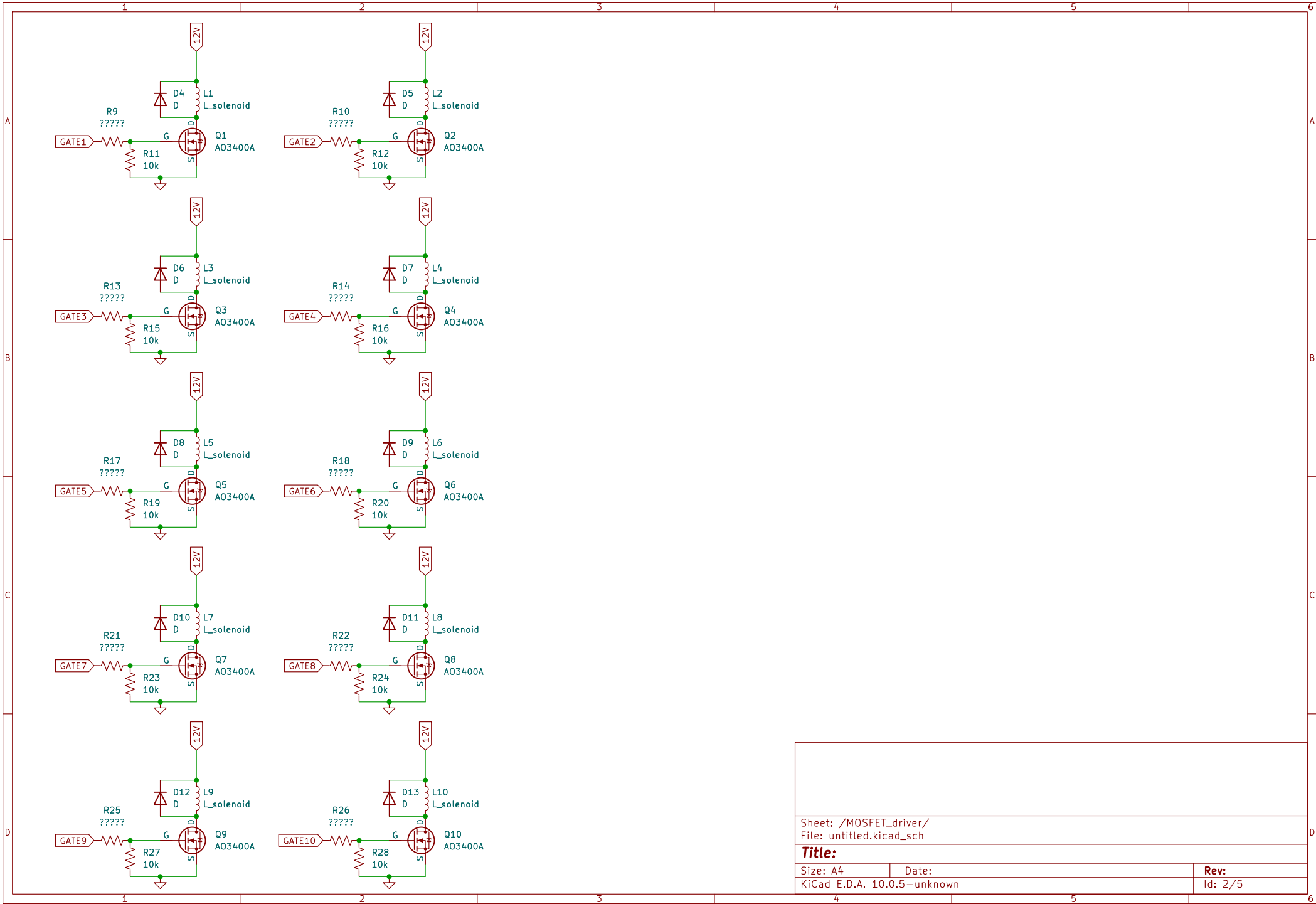
Sheet: /
File: X19-Control-Board.kicad_sch

Title: Purdue ROV Subsystem Board

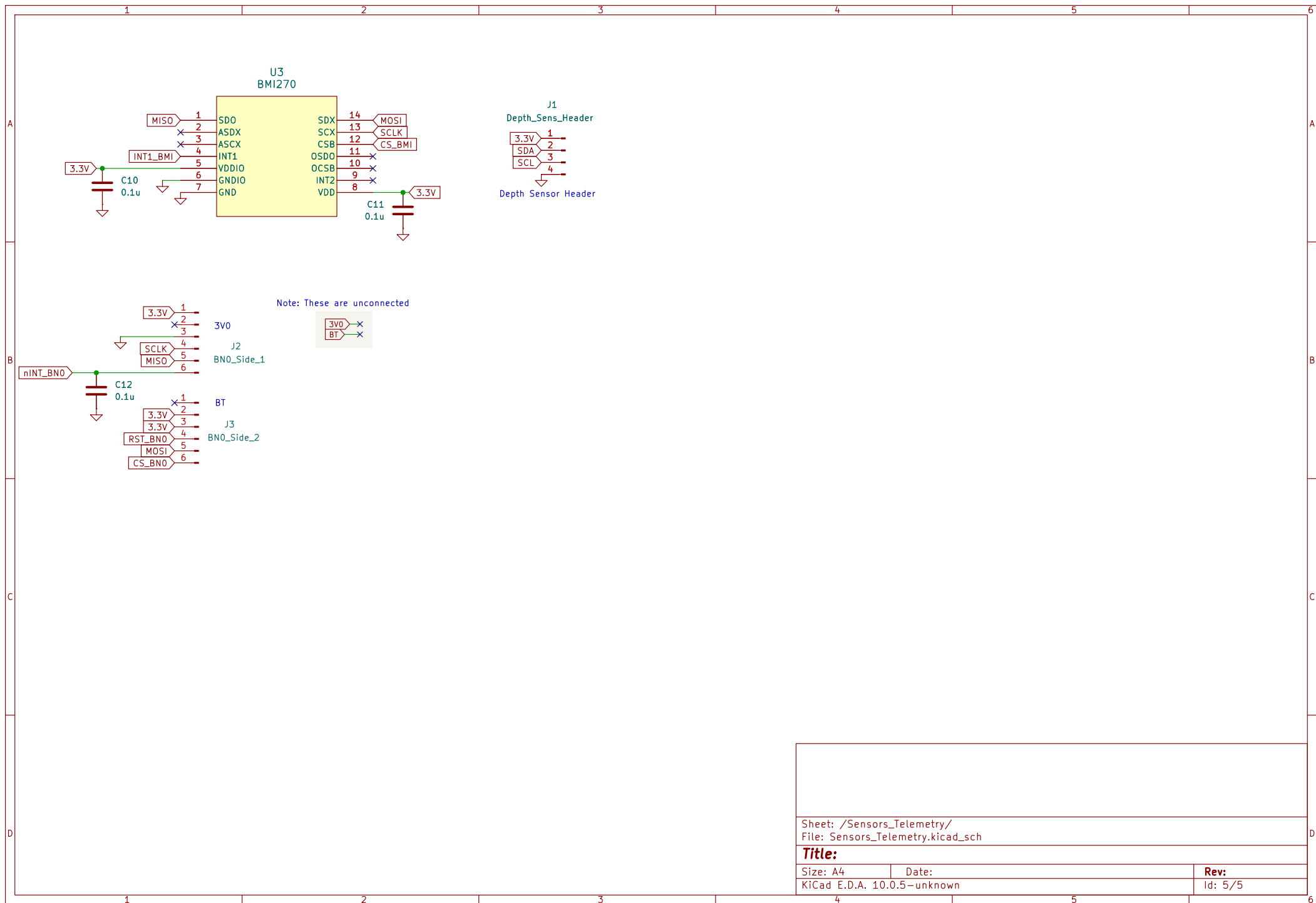
Size: A4
KiCad E.D.A. 10.0.5-unknown

Date:

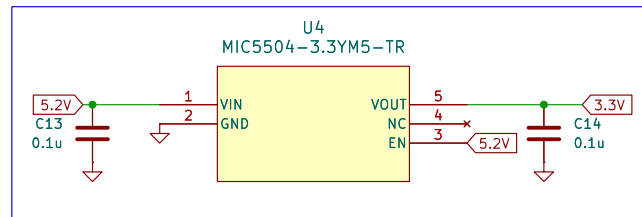
Rev:
Id: 1/5



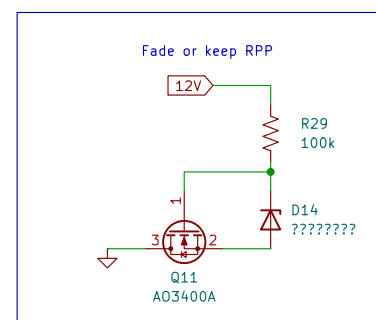
Sheet: /MOSFET_driver/ File: untitled.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5-unknown		Id: 2/5



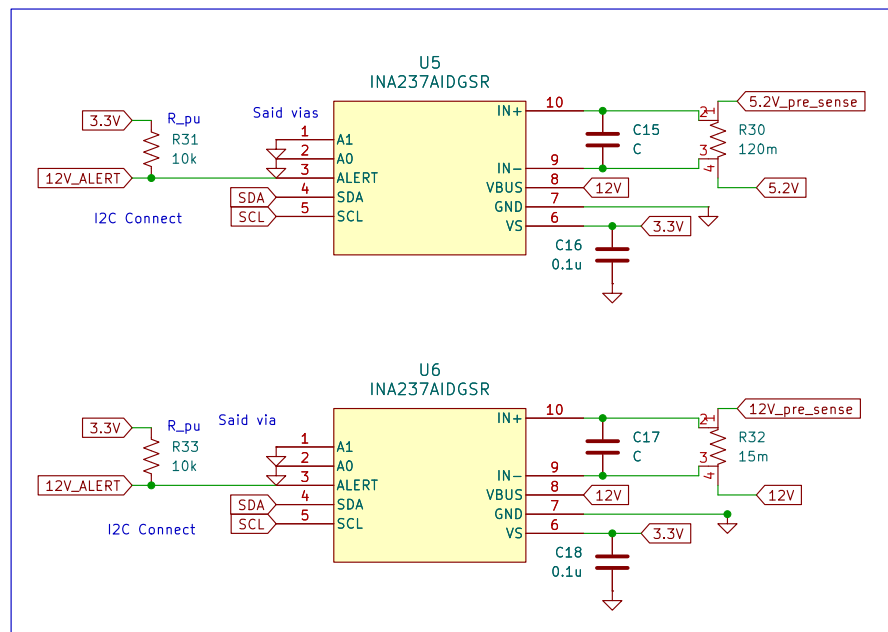
LDO – Voltage Conversion



Reverse Polarity Protection



Power Monitoring 5.2V, 12V rails



Sheet: /Voltage Conv-Protec-Monitoring/
File: Voltage_conv_prot.kicad_sch

Title:

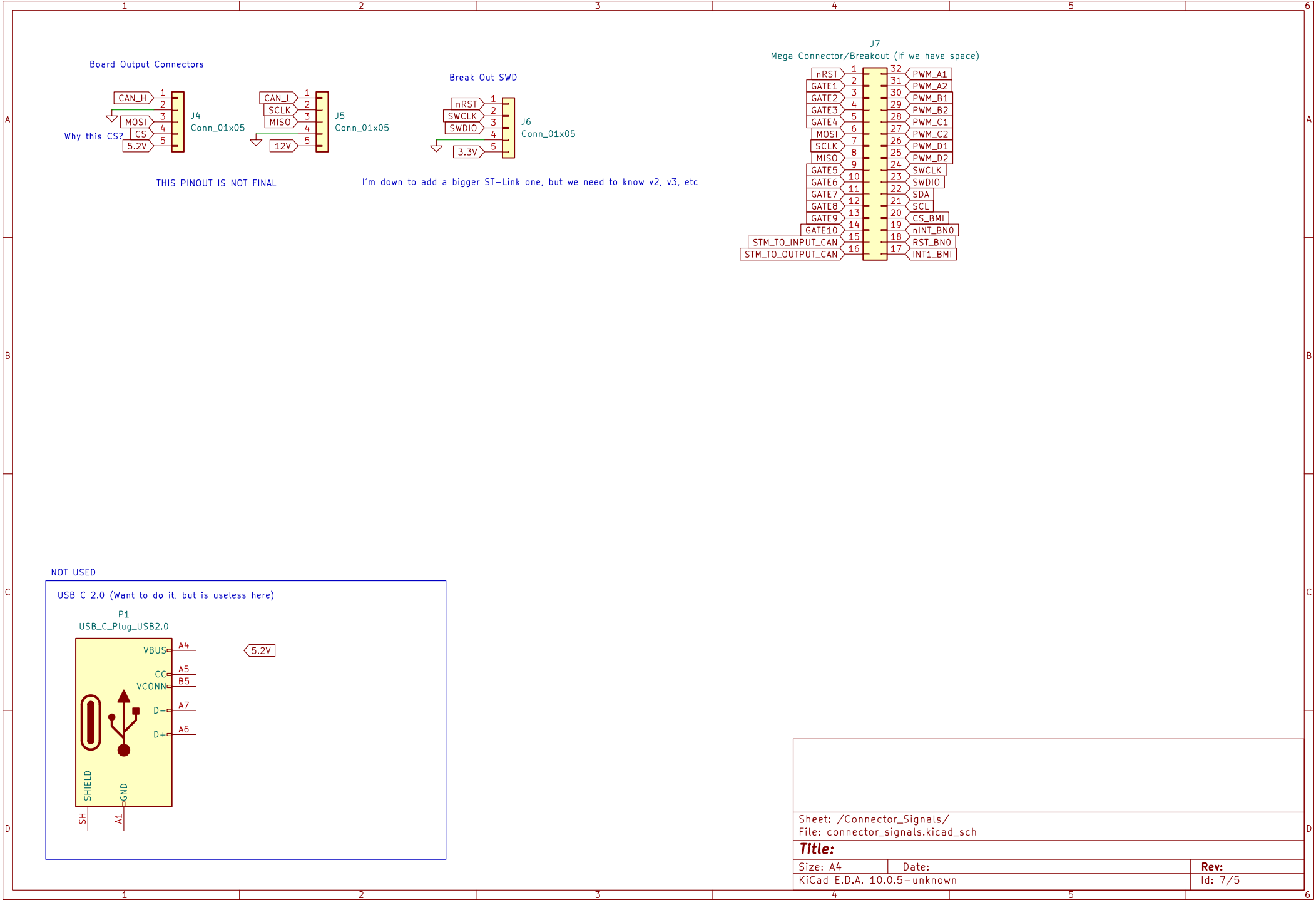
Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 6/5



THIS PINOUT IS NOT FINAL

I'm down to add a bigger ST-Link one, but we need to know v2, v3, etc

Break Out SWD

nRST

1

2

SWCLK

3

SWDIO

4

3.3V

5

J6

Conn_01x05

Mega Connector/Breakout (if we have space)

nRST

1

GATE1

2

GATE2

3

GATE3

4

GATE4

5

MOSI

6

SCLK

7

MISO

8

GATE5

9

GATE6

10

GATE7

11

GATE8

12

GATE9

13

GATE10

14

STM_TO_INPUT_CAN

15

STM_TO_OUTPUT_CAN

16

32

31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

PWM_A1

PWM_A2

PWM_B1

PWM_B2

PWM_C1

PWM_C2

PWM_D1

PWM_D2

SWCLK

SWDIO

SDA

SCL

CS_BMI

nINT_BN0

RST_BN0

INT1_BMI

NOT USED

USB C 2.0 (Want to do it, but is useless here)

P1

USB_C_Plug_USB2.0

VBUS

A4

CC

A5

VCONN

B5

D-

A7

D+

A6

SHIELD

SH

GND

A1

5.2V

Sheet: /Connector_Signals/
File: connector_signals.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 7/5